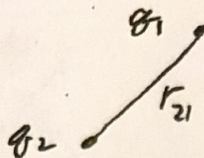


q_1



$$U_2 = U_{21} = \frac{k q_2 q_1}{r_{21}}$$

q_1

q_2

q_3

$$U_3 = U_{31} + U_{32} =$$

q_{100}

$$U_{100} = U_{100,1} + U_{100,2} + \dots + U_{100,99}$$

$$U_n = U_{n,1} + U_{n,2} + \dots + U_{n,n-1}$$

$$U_{\text{assembly}} = U_2 + U_3 + \dots + U_n$$

$$= (U_{21}) + (U_{31} + U_{32}) + \dots + (U_{n,1} + U_{n,2} + \dots + U_{n,n-1})$$

$$= U_{21} + \sum_{j=1}^2 U_{3j} + \dots + \sum_{j=1}^{n-1} U_{nj}$$

$$= \sum_{i=2}^n \sum_{j=1}^{i-1} U_{ij}$$